

IFEval Component-Wise Mixture Fit: $H=4$, $G=(3,3,1,3)$

This note summarizes a focused IFEval fit of the binary probit independent-mixture factor model with

```
H = 4
G = (3, 3, 1, 3)
lambda_l1_penalty = 4
pretraining iterations = 20 maximum
MAP refinement iterations = 20 maximum
```

The fitted full-data model stopped after 15 pretraining iterations and 12 MAP refinement iterations. The full-data binary probit log likelihood was -0.21897 per response. In the completed cross-validation table, this specification has held-out log likelihood -0.26978 per response, very close to the current best component-wise fit $G=(3,3,2,2)$, which has held-out log likelihood -0.26944 per response.

Component-Wise CV Context

The current component-wise CV screen strongly favors $H=4$ and $\lambda=4$. Among these fits, $G=(3,3,1,3)$ is near the top while keeping the third coordinate Gaussian.

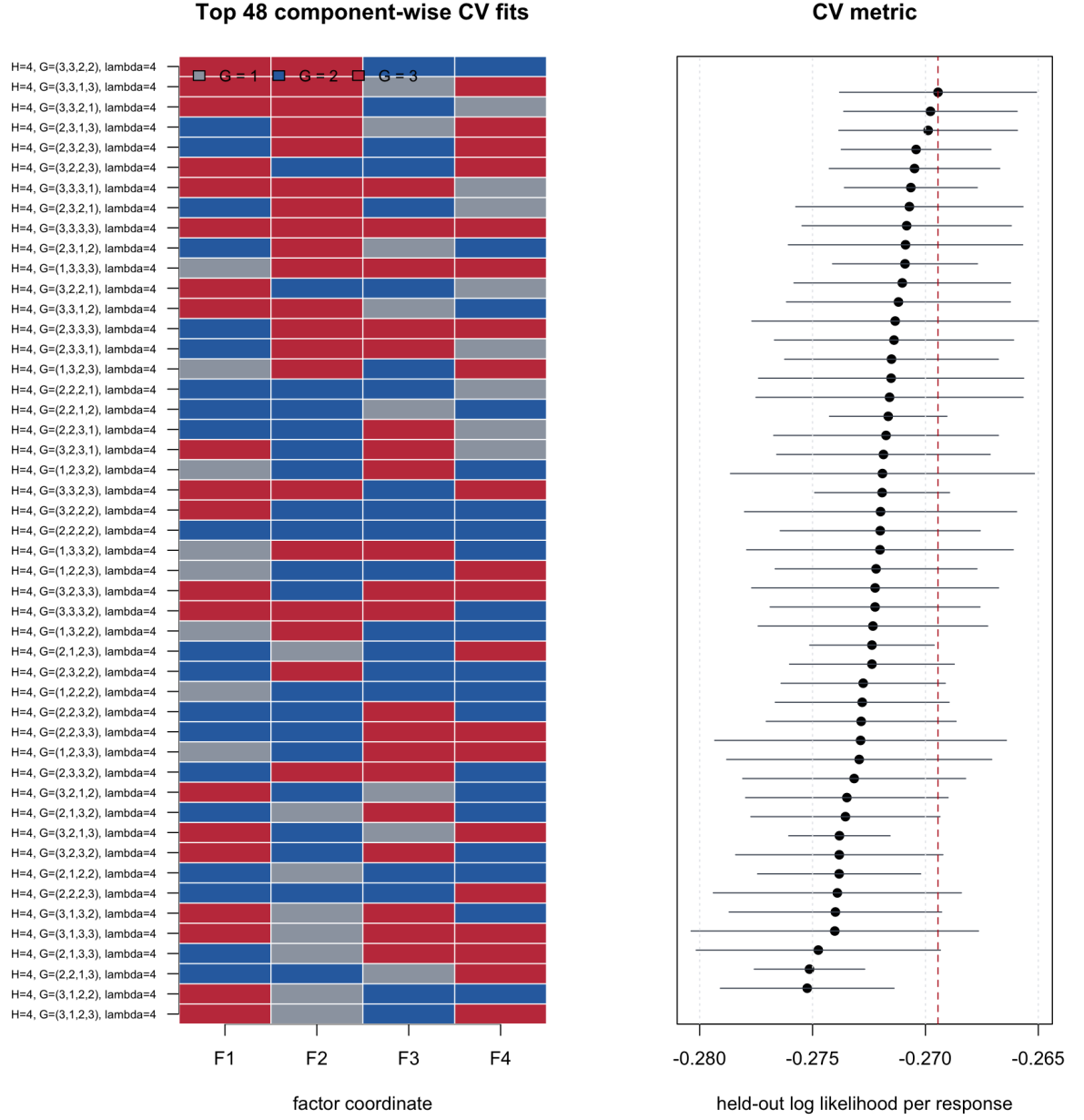


Figure 1: Component-wise CV metrics for $H=4$, $\lambda=4$

This is useful substantively: F3 is retained as a continuous axis, while F1, F2, and F4 are allowed to form discrete model subtypes.

Marginal Factor Distributions

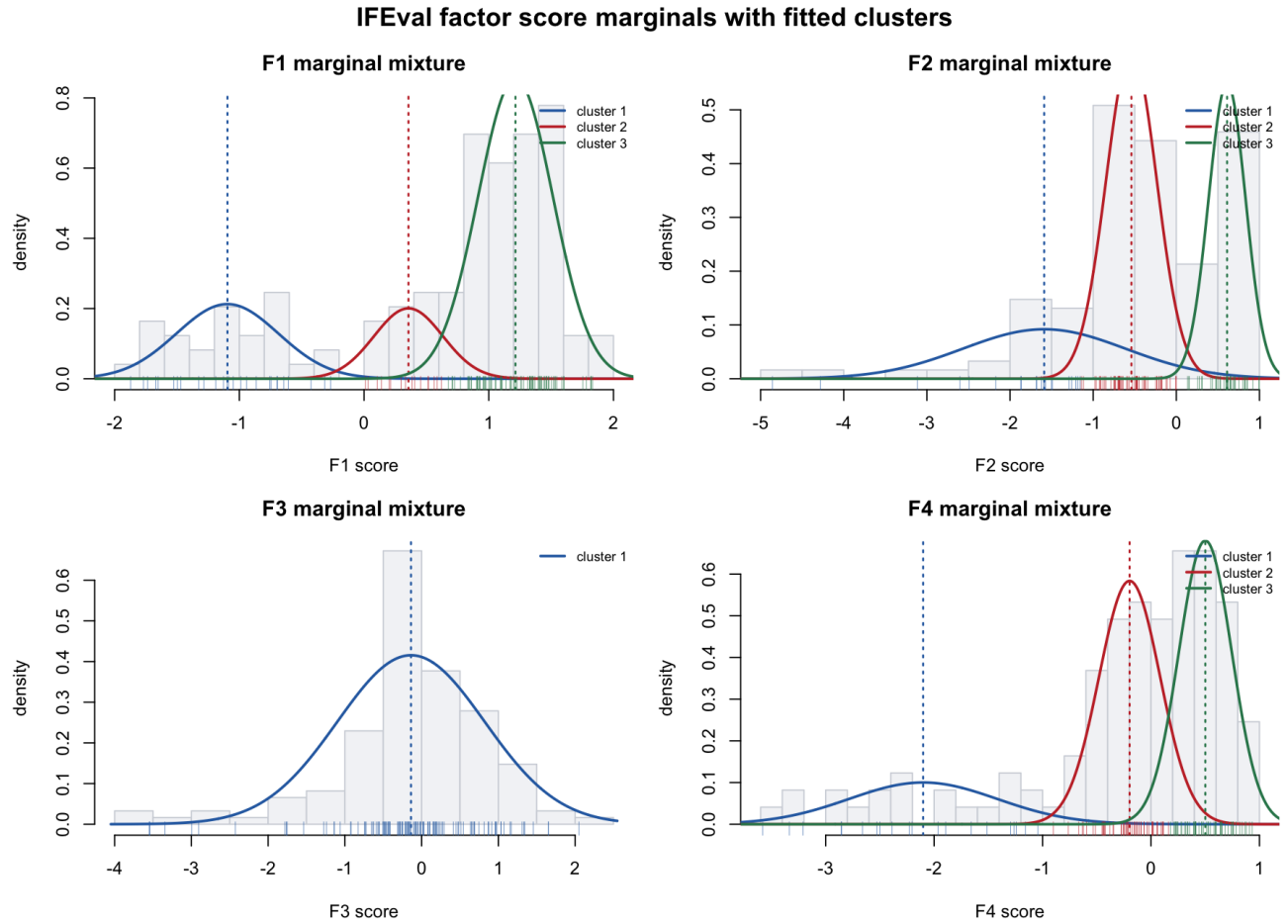


Figure 2: Marginal mixture fits

The fitted marginal mixture structure is:

Factor	Components	Interpretation
F1	3	Broad general instruction-following axis with high/mid/low model groups.
F2	3	A discrete axis for language/case/script constraints, especially all-lowercase and single-language requirements.
F3	1	A continuous axis rather than a discrete subtype. This is consistent with the earlier finding that extra F3 components mostly model central concentration rather than a meaningful left-tail group.

Factor	Components	Interpretation
F4	3	A discrete formatting/structured-output axis, including JSON wrapping, sentence-count limits, and local formatting constraints.

The fitted component sizes were:

Factor	Component counts
F1	27 / 16 / 79
F2	21 / 61 / 40
F3	122
F4	20 / 50 / 52

Loading Matrix

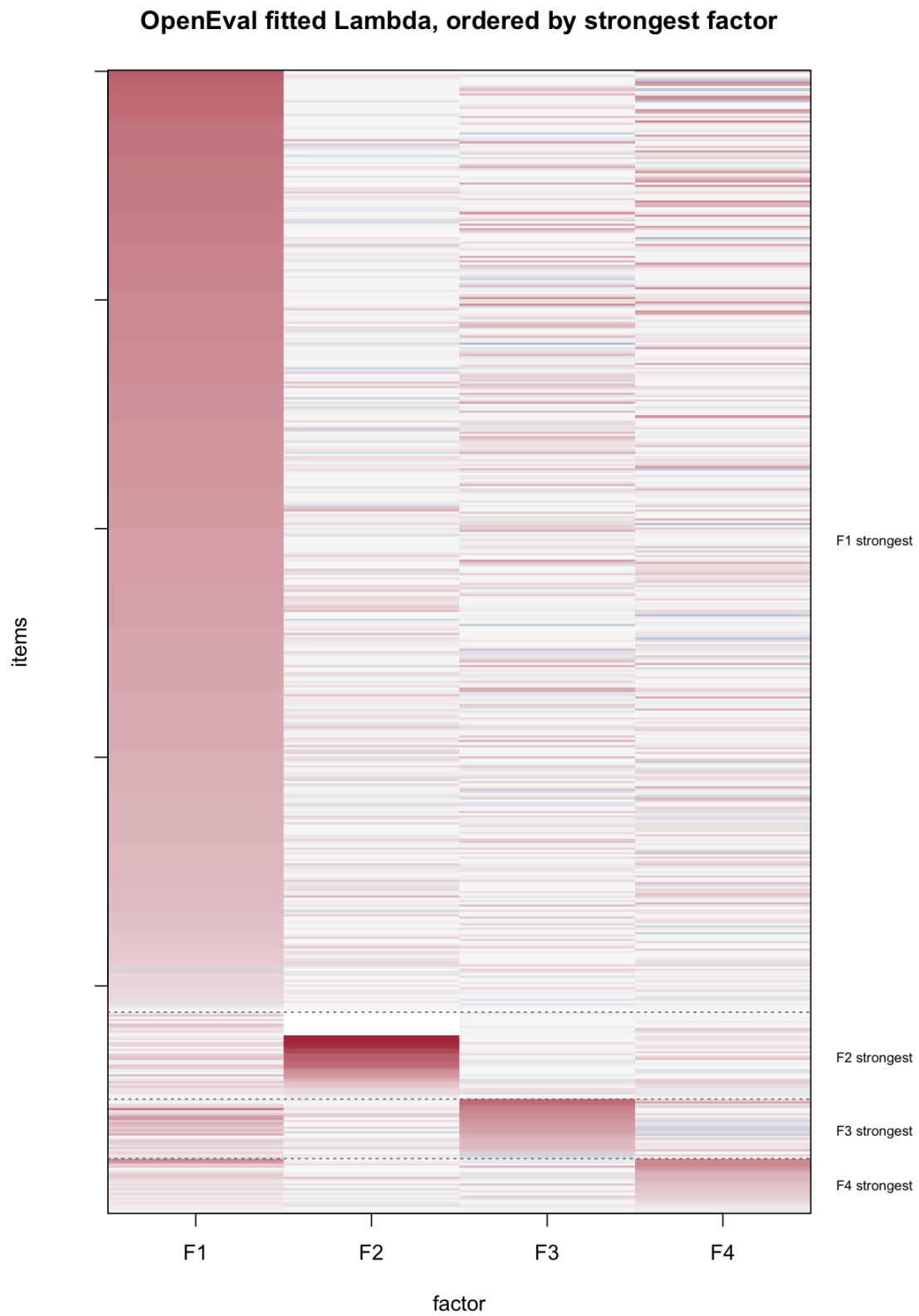


Figure 3: Lambda ordered by strongest factor

The heatmap is a row permutation of the fitted 500 x 4 loading matrix. For each item j , define

```
strongest_factor_j = argmax_h |lambda_jh|
max_abs_loading_j = max_h |lambda_jh|.
```

Rows are sorted by `strongest_factor_j`, then by decreasing `max_abs_loading_j`. Thus, the apparent blocks are not imposed by the model; they are a diagnostic ordering of the learned loadings. The fitted structure is dominated by a broad F1 factor, with smaller private-factor blocks for F2, F3, and F4.

Among active items with $|\text{loading}| > 0.1$, the strongest-factor counts are:

Strongest factor	Active items
F1	410
F2	37
F3	26
F4	23

The numeric loading summary also shows this structure:

Factor	Mean abs loading	$ \text{lambda} > 0.5$	$ \text{lambda} > 1.0$
F1	0.863	396	199
F2	0.240	38	28
F3	0.203	63	13
F4	0.222	69	18

Factor Interpretation

F1: Broad Instruction Following

F1 is the dominant general axis. High-loading items often combine ordinary task completion with explicit constraints, such as repeating the request exactly, ending with a required phrase, using all caps, or wrapping text in prescribed delimiters.

Representative high-F1 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_127	1.785	End the response with a specified sentence and no words after it.
ifeval_20260421T021146Z_242	1.758	Repeat the request exactly, then answer with a title in double angle brackets.
ifeval_20260421T021146Z_362	1.742	Repeat the request word-for-word before answering.
ifeval_20260421T021146Z_6	1.701	Write two labeled all-caps paragraphs.

F2: Language, Case, and Script Control

F2 is more specialized. The strongest positive F2 items emphasize lowercasing, single-language output, and constraints on allowed alphabets or scripts.

Representative high-F2 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_293	3.071	Product description entirely in lowercase.
ifeval_20260421T021146Z_409	2.967	English-only response, all lowercase.
ifeval_20260421T021146Z_280	2.963	Hindi poem, only Hindi, no other language.
ifeval_20260421T021146Z_292	2.822	Haiku in lowercase with no capital letters.

F3: Creative Lexical Constraint Axis

F3 is best treated as continuous in this fit. Its high-loading items often require creative generation under severe lexical constraints, such as avoiding or limiting a common letter while still producing a poem, song, quiz, or long essay.

Representative high-F3 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_277	1.637	Write a rubric as a poem while using the letter w fewer than two times.
ifeval_20260421T021146Z_477	1.565	Write a song while using the letter a at most once.
ifeval_20260421T021146Z_23	1.516	Write a logic quiz while using the letter t at most once.
ifeval_20260421T021146Z_147	1.324	Write a long essay of at least 50 sentences.

The Gaussian choice for F3 is important: it suggests this ability varies continuously across LLMs rather than forming a stable discrete subtype in the current data.

F4: Structured Output and Local Formatting

F4 captures a smaller but interpretable formatting axis. High-loading items frequently require JSON wrapping, sentence-count limits, required endings, highlighted sections, or other local structural constraints.

Representative high-F4 items include:

Item	Loading	Example constraint
ifeval_20260421T021146Z_269	1.219	Wrap the entire response in JSON format.
ifeval_20260421T021146Z_58	1.213	Answer in JSON format, with markdown ticks allowed.
ifeval_20260421T021146Z_359	1.207	Use fewer than 10 sentences and end with a required sentence.
ifeval_20260421T021146Z_33	1.173	Use a repeated keyword, fewer than six sentences, and highlighted text sections.

Working Interpretation

The $G=(3,3,1,3)$ fit gives a clean compromise between predictive performance and interpretability. F1 is the broad general IFEval ability. F2 and F4 describe discrete model subtypes for specialized constraint-following modes. F3 remains a continuous creative-control axis, especially for generative tasks with severe lexical restrictions.

This structure is more interpretable than forcing F3 to have multiple mixture components: when F3 is given two or three components, the extra components tend to model central concentration rather than a meaningful left-tail or high-skill subtype.